

Larry Nguyen

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Education

San Jose State University

Aug 2025 — May 2027

Masters of Science in Software Engineering

San Jose, CA

- **Relevant Coursework:** Software Systems Engineering, Enterprise Software Platforms, Enterprise App Development

University of California, San Diego

Sept 2021 — Jun 2025

Bachelors of Science in Machine Learning (GPA: 3.8)

La Jolla, CA

- **Relevant Coursework:** Data Science in Practice, Intro to Unsupervised Learning, AI Algorithms

Technical Skills

Languages: Java, Python, Javascript, C++, HTML, CSS, SQL

Technologies: React, Express.js, Node.js, MongoDB, AWS, Docker, Kubernetes, Pandas, Sklearn, Github Actions (CI/CD)

Concepts: Supervised Learning, Unsupervised Learning, Neural Networks, RESTful APIs, Full-Stack Development, Front-end, Backend, Microservices, Data Visualization, System Design, Git, CI/CD Pipelines, Unit Testing, Linux, Web Development, Object Oriented Design, Agile

Experience

JSR Micro Inc. Digital Technology Team

Jun 2023 — Sept 2023

Data Engineer Intern

Sunnyvale, CA

- Developed a custom ETL data pipeline via **Airbyte** integrating **OSI-PI Web API** and **Amazon Redshift** using **SQL**, organizing raw data into normalized, analysis-ready tables
- Optimized pipeline performance via **SQL** sort and distribution keys, reducing data ingestion time by 5 minutes across large-scale operational datasets exceeding 100K rows/records
- Extended pipeline functionality with multi-endpoint querying, time-windowed data pulls, and a configurable **REST API** settings interface

University of California, San Diego

Sept 2023 — Dec 2023

Computer Science Tutor

La Jolla, CA

- Conducted **code reviews** for 25+ students in Python, identifying performance bottlenecks and guiding refactors to improve readability, runtime, and **software design principles**
- Developed structured debugging frameworks adopted across 3 class sections, reducing repeated questions by standardizing common Python runtime and logic error patterns
- Taught Big O in the context of assignments, leading to a 75% increase in efficiency measured through benchmarking

Projects

Distributed System Queries | Python, C++, gRPC, Pybind

- Architected a **distributed systems** query service in **Python** and **C++** using **gRPC** with a leader/worker hierarchy to process 1M+ fire incident records with geospatial filtering
- Implemented a heartbeat system as an automatic failure detection between servers and reassignment logic to increase **fault tolerance** within the system and ensure task completion after server failure
- Improved throughput by 66% through horizontal scaling across 5 parallel workers, replacing single-node processing

Campus Marketplace | React, Javascript, Node.js, Mongo, Express, AWS EC2, NLP, LLM API

- Developed and deployed a **full-stack** web app via **AWS EC2**, enabling users to post listings, search items, and message sellers, with a **React** frontend and a **Node** backend, collaborating in a sprint-based Agile environment
- Integrated the **Google Gemini LLM API**, applying **prompt engineering** to translate natural-language queries into structured search filters using **NLP**
- Implemented a live messaging system with inbox management and per-listing threads, secured with **JWT authentication** and server-side access control; validated 15+ API endpoints through Postman integration testing

Trialize | Python, React Native, MongoDB, AWS Document DB

- Engineered a HIPAA-compliant mobile application using React Native, connecting patient data with clinical trial databases; identified 50+ potentially eligible trials per patient within initial testing
- Collaborated with a cross-functional team to design and document system architecture simulating SHARP's healthcare API using MongoDB for local testing prior to **AWS DocumentDB** migration for 1,000+ users